```
akansha@localhost:~$ pwd
/home/akansha
akansha@localhost:~$ ls
cd      Documents  linuxlab  Music      myfolder  practice  Public  Templates
Desktop  Downloads  linuxlB   myfirstlab  Pictures  project   rm.txt  Videos
akansha@localhost:~$ ls-l
bash: ls-l: command not found...
akansha@localhost:~$ cd
akansha@localhost:~$ pwd
/home/akansha
akansha@localhost:~$ ls -a
.            .bashrc  Documents  .mozilla    practice   Videos
..           .cache   Downloads  Music      project
.bash_history  cd      linuxlab   myfirstlab  Public
.bash_logout   .config  linuxlB    myfolder    rm.txt
.bash_profile  Desktop  .local     Pictures    Templates
akansha@localhost:~$ ls -lh
total 0
drwxr-xr-x. 2 akansha akansha  6 Oct  5 14:07 cd
drwxr-xr-x. 2 akansha akansha  6 Sep 21 15:52 Desktop
drwxr-xr-x. 2 akansha akansha  6 Sep 21 15:52 Documents
drwxr-xr-x. 2 akansha akansha  6 Sep 21 15:52 Downloads
drwxr-xr-x. 6 akansha akansha 87 Oct  5 14:55 linuxlab
drwxr-xr-x. 2 akansha akansha  6 Oct  5 14:07 linuxlB
drwxr-xr-x. 2 akansha akansha  6 Sep 21 15:52 Music
```

```
drwxr-xr-x. 2 akansha akansha 32 Oct 12 15:58 myfirstlab
drwxr-xr-x. 2 akansha akansha 39 Sep 21 22:06 myfolder
drwxr-xr-x. 2 akansha akansha  6 Sep 21 15:52 Pictures
drwxr-xr-x. 3 akansha akansha 22 Sep 22 20:20 practice
drwxr-xr-x. 3 akansha akansha 17 Oct 12 15:56 project
drwxr-xr-x. 2 akansha akansha  6 Sep 21 15:52 Public
-rw-r--r--. 1 akansha akansha  0 Sep 21 16:01 rm.txt
drwxr-xr-x. 2 akansha akansha  6 Sep 21 15:52 Templates
drwxr-xr-x. 2 akansha akansha  6 Sep 21 15:52 Videos
akansha@localhost:~$ mkdir myfirstlab
mkdir: cannot create directory ‘myfirstlab’: File exists
akansha@localhost:~$
cd myfirstlab
akansha@localhost:~/myfirstlab$ pwd
/home/akansha/myfirstlab
akansha@localhost:~/myfirstlab$ touch a.txt b.txt c.txt
akansha@localhost:~/myfirstlab$ ls
a.txt  b.txt  c.txt
akansha@localhost:~/myfirstlab$ cat /etc/os-release
NAME="Red Hat Enterprise Linux"
VERSION="10.0 (Coughlan)"
ID="rhel"
ID_LIKE="centos fedora"
VERSION_ID="10.0"
PLATFORM_ID="platform:el10"
```

PRETTY_NAME="Red Hat Enterprise Linux 10.0 (Coughlan)"

ANSI_COLOR="0;31"

LOGO="fedora-logo-icon"

CPE_NAME="cpe:/o:redhat:enterprise_linux:10::baseos"

HOME_URL="https://www.redhat.com/"

VENDOR_NAME="Red Hat"

VENDOR_URL="https://www.redhat.com/"

DOCUMENTATION_URL="https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/10"

BUG_REPORT_URL="https://issues.redhat.com/"

REDHAT_BUGZILLA_PRODUCT="Red Hat Enterprise Linux 10"

REDHAT_BUGZILLA_PRODUCT_VERSION=10.0

REDHAT_SUPPORT_PRODUCT="Red Hat Enterprise Linux"

REDHAT_SUPPORT_PRODUCT_VERSION="10.0"

akansha@localhost:~/myfirstlab$ whoami

akansha

akansha@localhost:~/myfirstlab$ hostname

localhost.localdomain

akansha@localhost:~/myfirstlab$ df -h

Filesystem Size Used Avail Use% Mounted on

/dev/mapper/rhel-root 17G 4.1G 13G 25% /

devtmpfs 4.0M 0 4.0M 0% /dev

tmpfs 836M 84K 836M 1% /dev/shm

tmpfs 335M 6.8M 328M 3% /run

tmpfs 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service

/dev/nvme0n1p2 960M 335M 626M 35% /boot

tmpfs 168M 144K 168M 1% /run/user/1000

akansha@localhost:~/myfirstlab$ free -h

total used free shared buff/cache available

Mem: 1.6Gi 1.2Gi 227Mi 11Mi 334Mi 395Mi

Swap: 2.0Gi 105Mi 1.9Gi

akansha@localhost:~/myfirstlab$ uptime

19:06:52 up 27 min, 2 users, load average: 0.67, 0.37, 0.33

akansha@localhost:~/myfirstlab$ uname -a

Linux localhost.localdomain 6.12.0-55.9.1.el10_0.x86_64 #1 SMP PREEMPT_DYNAMIC Tue Mar 25 09:14:09 EDT 2025 x86_64 GNU/Linux

akansha@localhost:~/myfirstlab$ man ls

akansha@localhost:~/myfirstlab$ whatis mv

mv: nothing appropriate.

akansha@localhost:~/myfirstlab$ cd ~

akansha@localhost:~$ mkdir -p project/src/data

akansha@localhost:~$ ls -R project

project:

src

project/src:

data

project/src/data:

akansha@localhost:~$ cd ~

akansha@localhost:~$ rm myfirstlab/c.txt

```
akansha@localhost:~$ ls myfirstlab

a.txt  b.txt

akansha@localhost:~$ cd ~

akansha@localhost:~$ echo "Akansha" > students.txt

echo "Juhi" >> students.txt

echo "Shristi" >> students.txt

echo "Ayushi" >> students.txt

echo "Priya" >> students.txt

akansha@localhost:~$ cat students.txt

Akansha

Juhi

Shristi

Ayushi

Priya

akansha@localhost:~$ head -n 2 students.txt

Akansha

Juhi

akansha@localhost:~$ tail -n 2 students.txt

Ayushi

Priya

akansha@localhost:~$ mkdir -p backup

cp students.txt backup/

akansha@localhost:~$ ls backup

students.txt

akansha@localhost:~$ mv students.txt class_list.txt
```

```
akansha@localhost:~$ ls
backup        Desktop   linuxlab  myfirstlab  practice  rm.txt
cd            Documents linuxlB   myfolder    project   Templates
class_list.txt  Downloads  Music    Pictures    Public    Videos
akansha@localhost:~$ grep 'a' class_list.txt
Akansha
Priya
akansha@localhost:~$ grep -n 'a' class_list.txt
1:Akansha
5:Priya
akansha@localhost:~$ chmod 600 class_list.txt
akansha@localhost:~$ ls -l class_list.txt
-rw-------. 1 akansha akansha 34 Oct 21 19:13 class_list.txt
akansha@localhost:~$ -rw-------
bash: -rw-------: command not found...
akansha@localhost:~$ find ~ -type f -name "*.txt"
/home/akansha/rm.txt
/home/akansha/myfolder/hello.txt
/home/akansha/myfolder/copy.txt
/home/akansha/practice/practice/hello.txt
/home/akansha/linuxlab/docs/notes.txt
/home/akansha/linuxlab/docs/nots.txt
/home/akansha/linuxlab/docs/tasks.txt
/home/akansha/linuxlab/data/employee.txt
/home/akansha/linuxlab/data/empoyee.txt
```

/home/akansha/linuxlab/data/report.txt

/home/akansha/linuxlab/backup/empoyee.txt

/home/akansha/linuxlab/backup/docs_copy/report.txt

/home/akansha/linuxlab/backup/docs_copy/todo.txt

/home/akansha/linuxlab/backup/docs_copy/nots.txt

/home/akansha/linuxlab/backup/docs_copy/notes.txt

/home/akansha/linuxlab/commands_history.txt

/home/akansha/myfirstlab/a.txt

/home/akansha/myfirstlab/b.txt

/home/akansha/backup/students.txt

/home/akansha/class_list.txt

akansha@localhost:~$ wc -l class_list.txt

5 class_list.txt

akansha@localhost:~$ head -n 5 /etc/passwd

root:x:0:0:Super User:/root:/bin/bash

bin:x:1:1:bin:/bin:/usr/sbin/nologin

daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin

adm:x:3:4:adm:/var/adm:/usr/sbin/nologin

lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin

akansha@localhost:~$ sudo tail -n 5 /var/log/messages

[sudo] password for akansha:

Oct 21 19:27:44 localhost systemd[1]: Starting user-runtime-dir@0.service - User Runtime Directory /run/user/0...

Oct 21 19:27:44 localhost systemd-logind[910]: New session c3 of user root.

Oct 21 19:27:44 localhost systemd[1]: Finished user-runtime-dir@0.service - User Runtime Directory /run/user/0.

```
Oct 21 19:27:44 localhost systemd[1]: Starting user@0.service - User Manager for UID 0...
Oct 21 19:27:44 localhost systemd-logind[910]: New session 4 of user root.
akansha@localhost:~$ ls /var/log
anaconda         lastlog          tuned
audit            maillog          vmware-network.1.log
boot.log         maillog-20261005   vmware-network.2.log
boot.log-20261005  maillog-20261016   vmware-network.3.log
boot.log-20261016  messages         vmware-network.4.log
boot.log-20261021  messages-20261005  vmware-network.5.log
btmp             messages-20261016  vmware-network.6.log
btmp-20261005      private          vmware-network.7.log
chrony           qemu-ga          vmware-network.8.log
cron             rhsm             vmware-network.9.log
cron-20261005      samba            vmware-network.log
cron-20261016      secure           vmware-vgauthsvc.log.0
cups             secure-20261005    vmware-vmsvc-root.log
dnf.librepo.log    secure-20261016    vmware-vmtoolsd-akansha.log
dnf.log          speech-dispatcher  vmware-vmtoolsd-root.log
dnf.rpm.log        spooler          vmware-vmusr-akansha.log
firewalld        spooler-20261005   wtmp
gdm              spooler-20261016
insights-client    sssd
akansha@localhost:~$ cp -r project project_backup
akansha@localhost:~$ ls -R project_backup
project_backup:
```

src

project_backup/src:

data

project_backup/src/data:

akansha@localhost:~$ less /etc/passwd

akansha@localhost:~$ mkdir -p archive

akansha@localhost:~$ mv myfirstlab/a.txt myfirstlab/b.txt archive/

akansha@localhost:~$ ls archive

a.txt b.txt

akansha@localhost:~$ mkdir -p logs

for i in {1..10}; do echo "Log entry $i"; done > logs/app.log

akansha@localhost:~$ tail -n 3 logs/app.log

Log entry 8

Log entry 9

Log entry 10

akansha@localhost:~$ find . -type f -name "*.log" -exec cat {} \;

��jour�y����jour�Y��Log entry 1

Log entry 2

Log entry 3

Log entry 4

Log entry 5

Log entry 6

Log entry 7

Log entry 8

Log entry 9

Log entry 10

akansha@localhost:~$ find . -type f -name "*.log" -exec sh -c 'echo "File: $1"; cat "$1"' _ {} \;

File: ./.local/share/gvfs-metadata/home-14af7986.log

��jour�y��File: ./.local/share/gvfs-metadata/root-b45910eb.log

��jour�Y��File: ./logs/app.log

Log entry 1

Log entry 2

Log entry 3

Log entry 4

Log entry 5

Log entry 6

Log entry 7

Log entry 8

Log entry 9

Log entry 10

akansha@localhost:~$ grep -ri "error" .

grep: ./.cache/gstreamer-1.0/registry.x86_64.bin: binary file matches

grep: ./.cache/gnome-software/appstream/components.xmlb: binary file matches

./.cache/gnome-software/odrs/ratings.json: "info.urbanterror.UrbanTerror": {

./.cache/gnome-software/odrs/ratings.json: "io.snapcraft.dunes-of-terror-2w5xWxwrwGekXcW1AiQwwq3dg53Tr6RC": {

./.cache/gnome-software/odrs/ratings.json: "io.snapcraft.urban-terror-wyD87SBPoJn19vtXkUq30CHOHWi0LFF3": {

./.cache/gnome-software/odrs/ratings.json: "urbanterror.desktop": {

```
akansha@localhost:~$ cat > test.sh <<'EOF'
#!/bin/bash
# rwxr-xr-- means owner=rwx, group=r-x, others=r--
echo "Hello from test.sh"
EOF
akansha@localhost:~$ chmod +x test.sh
akansha@localhost:~$ ls -l test.sh
-rwxr-xr-x. 1 akansha akansha 89 Oct 21 19:35 test.sh
akansha@localhost:~$ ./test.sh
Hello from test.sh
akansha@localhost:~$ chmod 754 test.sh
akansha@localhost:~$ mkdir demo && cd demo && ls
akansha@localhost:~/demo$ mkdir -p demo && cd demo && ls
akansha@localhost:~/demo/demo$ find ~ -type f -name "*.txt" | wc -l
20
akansha@localhost:~/demo/demo$ find /var/log -type d
/var/log
/var/log/private
find: ‘/var/log/private’: Permission denied
/var/log/samba
find: ‘/var/log/samba’: Permission denied
/var/log/audit
find: ‘/var/log/audit’: Permission denied
/var/log/speech-dispatcher
find: ‘/var/log/speech-dispatcher’: Permission denied
```

/var/log/sssd

find: ‘/var/log/sssd’: Permission denied

/var/log/cups

/var/log/rhsm

/var/log/insights-client

find: ‘/var/log/insights-client’: Permission denied

/var/log/tuned

/var/log/gdm

find: ‘/var/log/gdm’: Permission denied

/var/log/chrony

find: ‘/var/log/chrony’: Permission denied

/var/log/qemu-ga

/var/log/anaconda

akansha@localhost:~/demo/demo$ grep -i "UUID" /etc/fstab

UUID=723de731-6879-4100-bd0c-8e4401951fde / xfs defaults 0 0

UUID=ae03e04f-40cc-4b5b-b6a1-772c29979393 /boot xfs defaults 0 0

UUID=d7fc25e6-e859-4c73-8c89-e1653a2fae27 none swap defaults 0 0

akansha@localhost:~/demo/demo$ du -sh project

du: cannot access 'project': No such file or directory

akansha@localhost:~/demo/demo$ ls -lS /var/log | head -n 4

total 4380

-rw-------. 1 root root 3220468 Oct 5 14:06 messages-20261005

-rw-------. 1 root root 363918 Oct 5 14:06 boot.log-20261005

-rw-------. 1 root root 310896 Oct 21 19:40 messages

akansha@localhost:~/demo/demo$ find /var/log -maxdepth 1 -type f -printf '%s %p\n' | sort -nr | head -n 3

3220468 /var/log/messages-20261005

363918 /var/log/boot.log-20261005

311464 /var/log/messages

akansha@localhost:~/demo/demo$ pwd

ls -la

/home/akansha/demo/demo

total 0

drwxr-xr-x. 2 akansha akansha 6 Oct 21 19:36 .

drwxr-xr-x. 3 akansha akansha 18 Oct 21 19:36 ..

akansha@localhost:~/demo/demo$ touch class_list.txt

akansha@localhost:~/demo/demo$ printf "Apple\nMango\nApple\nBanana\nMango\n" > fruits.txt

akansha@localhost:~/demo/demo$ sort fruits.txt | uniq

Apple

Banana

Mango

akansha@localhost:~/demo/demo$ uname -a > sys_info.txt

akansha@localhost:~/demo/demo$ cat sys_info.txt

Linux localhost.localdomain 6.12.0-55.9.1.el10_0.x86_64 #1 SMP PREEMPT_DYNAMIC Tue Mar 25 09:14:09 EDT 2025 x86_64 GNU/Linux

akansha@localhost:~/demo/demo$ find /var/log -type f -mtime -7

find: ‘/var/log/private’: Permission denied

/var/log/wtmp

/var/log/lastlog

find: ‘/var/log/samba’: Permission denied

find: ‘/var/log/audit’: Permission denied

find: ‘/var/log/speech-dispatcher’: Permission denied

find: ‘/var/log/sssd’: Permission denied

/var/log/rhsm/rhsmcertd.log

/var/log/rhsm/rhsm.log

/var/log/rhsm/rhsmcertd.log-20261016

/var/log/rhsm/rhsm.log-20261016

find: ‘/var/log/insights-client’: Permission denied

/var/log/tuned/tuned.log

find: ‘/var/log/gdm’: Permission denied

find: ‘/var/log/chrony’: Permission denied

/var/log/boot.log

/var/log/vmware-vgauthsvc.log.0

/var/log/vmware-vmtoolsd-root.log

/var/log/vmware-vmsvc-root.log

/var/log/vmware-vmtoolsd-akansha.log

/var/log/vmware-vmusr-akansha.log

/var/log/boot.log-20261016

/var/log/maillog

/var/log/secure

/var/log/vmware-network.log

/var/log/dnf.log

/var/log/dnf.rpm.log

/var/log/boot.log-20261021

/var/log/messages

/var/log/spooler

/var/log/dnf.librepo.log

/var/log/cron

/var/log/messages-20261016

/var/log/vmware-network.1.log

akansha@localhost:~/demo/demo$ sudo find /var/log -type f -mtime -7

[sudo] password for akansha:

/var/log/wtmp

/var/log/lastlog

/var/log/audit/audit.log

/var/log/sssd/sssd_kcm.log-20261016

/var/log/sssd/sssd_kcm.log

/var/log/rhsm/rhsmcertd.log

/var/log/rhsm/rhsm.log

/var/log/rhsm/rhsmcertd.log-20261016

/var/log/rhsm/rhsm.log-20261016

/var/log/tuned/tuned.log

/var/log/boot.log

/var/log/vmware-vgauthsvc.log.0

/var/log/vmware-vmtoolsd-root.log

/var/log/vmware-vmsvc-root.log

/var/log/vmware-vmtoolsd-akansha.log

/var/log/vmware-vmusr-akansha.log

/var/log/boot.log-20261016

/var/log/maillog

/var/log/secure

/var/log/vmware-network.log

/var/log/dnf.log

/var/log/dnf.rpm.log

/var/log/boot.log-20261021

/var/log/messages

/var/log/spooler

/var/log/dnf.librepo.log

/var/log/cron

/var/log/messages-20261016

/var/log/vmware-network.1.log

akansha@localhost:~/demo/demo$

akansha@localhost:~/demo/demo$ ^C

akansha@localhost:~/demo/demo$